

Movie Recommendation System

Developed a CLI-based Movie Recommendation System using Machine Learning techniques such as Collaborative Filtering and Content-Based Filtering. The system recommends movies based on user preferences, ratings, genres, and tags. Implemented cosine similarity, TF-IDF vectorization, and user-item similarity matrices to generate accurate movie recommendations.

The project supports:

- * Movie-based recommendations
- * User-based recommendations
- * Top-N recommended movie suggestions
- * Content analysis using genres and tags

Technologies used include Python, Pandas, NumPy, and Scikit-learn for data processing, similarity computation, and recommendation modeling.

Key learning outcomes:

- * Recommendation system algorithms
- * Cosine similarity and Pearson correlation
- * TF-IDF and basic NLP techniques
- * User-item matrix creation
- * Data preprocessing with Pandas and NumPy
- * Real-world machine learning workflow